

Lawrence Hollom

Institute of Mathematics, EPFL SB MATH, 1015 Lausanne, Switzerland

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Current Position

Postdoctoral Researcher

Supervisor: Marius Tiba

February 2026 – Present

EPFL

Research Interests

- Infinitary and structural combinatorics
- Extremal and probabilistic combinatorics
- Random graphs and percolation
- High-dimensional geometry

Education

PhD in Pure Mathematics

Supervisor: Béla Bollobás

September 2022 – January 2026

University of Cambridge

Part III of the Mathematical Tripos

Master's degree

September 2019 – July 2020

University of Cambridge

Bachelor's degree in Mathematics

First class, ranked 2nd in year

September 2016 – July 2019

University of Cambridge

Awards

Foundation of Science Award

ICBS, 2026

Smith–Knight and Raleigh–Knight First Prize

University of Cambridge, 2024

Previous roles

Jane Street Capital

Quantitative Trader

September 2020 – January 2022

London, UK

Invited Talks

2025 October

Combinatorics Seminar, University of Cambridge

2025 February

Discrete Mathematics Seminar, London School of Economics

2025 January

Combinatorics Seminar, University College London

2024 May

Combinatorics Seminar, University of Cambridge

Teaching

Teaching assistant Convex Geometric Analysis, Spring semester

EPFL, 2026

Supervisions Part II Logic and Set Theory, Lent term

University of Cambridge, 2025

Supervisions Part IA Numbers and Sets, Michaelmas term

University of Cambridge, 2024

Supervisions Part II Logic and Set Theory, Lent term

University of Cambridge, 2024

Supervisions Part IA Numbers and Sets, Michaelmas term

University of Cambridge, 2023

Supervisions Part IB Analysis and Topology, Michaelmas term

University of Cambridge, 2022

Technical skills

Programming Languages/Tools Languages

Rust, Lean, OCaml, Java, Python, C#, L^AT_EX
English (native), French (intermediate)

Publications and preprints

- Uniformly balanced H -factors in multicoloured complete graphs** 2026
[arXiv preprint](#)
with A. Banerjee
- Counterexamples to conjectures on strong maximality and minimality** 2026
[Electronic Journal of Combinatorics](#)
with B. Randall Shaw
- Reverse Littlewood–Offord problems with parity conditions** 2025
[arXiv preprint](#)
with G. B. Sorkin
- Connecting hypercube 1-factors** 2025
[arXiv preprint](#)
with B. Randall Shaw
- Approximate Itai-Zehavi conjecture for random graphs** 2025
[arXiv preprint](#)
with L. Lichev, A. Mond, J. Portier, and Y. Wang
- Finding a long cycle in a percolated vertex expander graph** 2025
[Random Structures & Algorithms](#)
- Monotonicity and decompositions of random regular graphs** 2025
[arXiv preprint](#) (Annals of Applied Probability, accepted)
with L. Lichev, A. Mond, J. Portier, and Y. Wang
- Double-jump phase transition for the reverse Littlewood–Offord problem** 2025
[arXiv preprint](#) (Journal of the London Mathematical Society, accepted)
with J. Portier and V. Souza
- The Aharoni–Korman conjecture is false** 2025
[arXiv preprint](#) (Israel Journal of Mathematics, accepted)
- A note on high-dimensional discrepancy of subtrees** 2024
[arXiv preprint](#) (Discrete & Computational Geometry, accepted)
with L. Lichev, A. Mond, and J. Portier
- Discrepancies of spanning trees in dense graphs** 2024
[arXiv preprint](#)
with L. Lichev, A. Mond, and J. Portier
- A uniform bound on almost colour-balanced perfect matchings in colour-balanced complete graphs** 2025
[Electronic Journal of Combinatorics](#)
- Almost colour-balanced spanning forests in complete graphs** 2024
[arXiv preprint](#)
with A. Mond and J. Portier
- The bunkbed conjecture is not robust to generalisation** 2025
[European Journal of Combinatorics](#)

On graphs with maximum difference between game chromatic number and chromatic number 2025

Discrete Mathematics

On monotonicity in Maker-Breaker graph colouring games 2024

Discrete Applied Mathematics

Tight Anti-Concentration of Rademacher Sums 2025

Random Structures & Algorithms

with J. Portier

A note on interval colourings of graphs 2024

European Journal of Combinatorics

with M. Axenovich, A. Girão, J. Portier, E. Powierski, M. Savery, Y. Tamitegama, and L. Versteegen

A new proof of the bunkbed conjecture in the $p \uparrow 1$ limit 2024

Discrete Mathematics